

ASSIGNMENT 2

Sensors and Actuators in Computational Hardware

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Introduction

Sensors and actuators form the backbone of computational hardware, enabling interaction with the physical world through sensing motion and producing precise movements. The gyroscope sensor detects angular changes, while the harmonic drive actuator delivers high-precision torque. These components are essential in Arduino-based projects and advanced robotics, supporting stability and control. This report explains their definition, working principle, key features, and real-world applications.

Gyroscope Sensor

Definition



A gyroscope sensor measures angular velocity and maintains orientation by detecting rotational motion across three axes. Modern versions use MEMS technology for compact integration and are widely used in embedded systems.

Construction

A typical MEMS gyroscope consists of a vibrating proof mass, drive frame, sense electrodes, and signal processing circuitry. The proof mass vibrates at a fixed frequency, and when rotational motion occurs, the Coriolis force causes displacement that is detected by capacitive sensors.

Working Principle

A gyroscope sensor operates based on the Coriolis effect. A vibrating proof mass deflects when the device rotates, which changes capacitance between fixed and moving plates. The electronics convert this change into angular rate data, enabling real-time orientation



tracking.

Key Features

- High sensitivity to angular motion
- Three-axis rotation measurement
- Low power consumption
- Compact MEMS design
- Easy microcontroller interfacing (I2C/SPI)

Advantages

- Small size and lightweight
- High reliability
- Fast response time
- Low power usage
- Suitable for portable devices

Disadvantages

- Sensitive to temperature drift
- Requires calibration
- Noise in low-cost sensors
- Limited long-term accuracy

Applications

Gyroscope sensors are used in smartphones for screen rotation, drones for flight stabilization, robotics for navigation, and automotive systems for electronic stability control.



Harmonic Drive Actuator



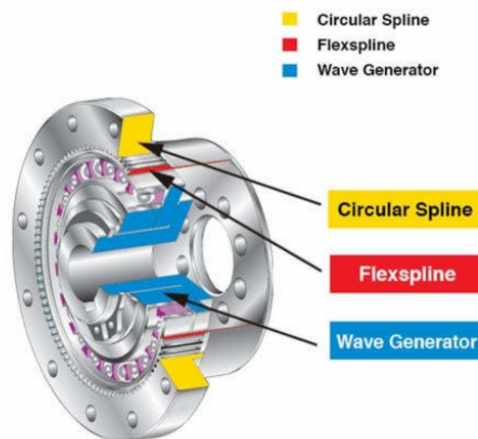
Definition

A harmonic drive actuator is a strain wave gear system that provides zero backlash and high reduction ratio in a compact structure. It is widely used for precision motion control applications.

Main components

- Wave Generator
- Flexspline
- Circular Spline

Working Principle



The harmonic drive consists of three main components: wave generator, flexspline, and circular spline. The wave generator deforms the flexspline into an elliptical shape, causing

progressive tooth engagement with the circular spline. This results in reduced output speed with high torque and precise motion.

Key Features

- Zero backlash for high precision
- High reduction ratio (50:1 to 160:1)
- Compact and lightweight
- High torque-to-weight ratio
- Smooth and accurate positioning

Advantages

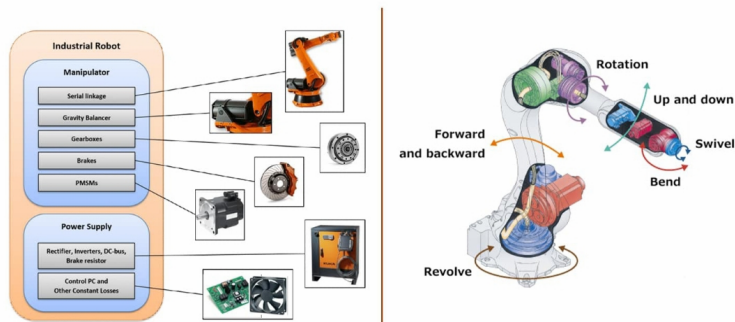
- High precision positioning
- Lightweight design
- Compact structure
- High torque-to-weight ratio
- No backlash

Disadvantages

- Expensive compared to normal gears
- Limited speed range
- Requires careful alignment
- Flexspline fatigue over long usage

Applications

Harmonic drive actuators are used in industrial robot arms, humanoid robots, CNC machines, aerospace systems, and medical devices requiring accurate positioning and repeatability.



Conclusion

The gyroscope sensor and harmonic drive actuator enhance computational hardware systems by providing reliable motion sensing and precision actuation. Their integration in robotics, consumer electronics, and automation demonstrates their importance in modern engineering. These components enable stability, accuracy, and control, paving the way for advanced automation and smart devices.